

Outlier Sensitivity of Envelope TSCP

Controlled rf2 and Crime experiments
with training-target standardization

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Abstract

We investigate whether a single extreme observation explains the loss of envelope TSCP to Point CHR in existing multi-output regression experiments. In rf2, 200 matched random splits with training-target standardization show that deleting the isolated NASI2 value 78.9 reduces envelope’s mean full prediction volume by 85.55%, with essentially unchanged joint coverage. The envelope/CHR ratio of mean volumes decreases from 6.347 to 1.477; the ranking does not reverse. A screen of eight existing cohorts identifies Crime as a second candidate. Removing its largest-population community changes the standardized ratio from 38.615 to 0.101. This numerical reversal is driven by rare extreme-volume calibration splits: after deletion, envelope is smaller in 100 of 200 splits, and a descriptive bootstrap interval for the mean-volume ratio includes one. The evidence supports sensitivity to influential observations, while distinguishing a reversal in the empirical mean from a consistent advantage across splits. Neither deletion establishes that the removed record is erroneous.

1 Experimental design and estimands

The primary questions are whether rf2’s isolated extreme observation remains influential after improving the location model’s target weighting, and whether another existing real-data benchmark exhibits a loss-to-win reversal against Point CHR. Throughout, “TSCP” means the current surface-envelope construction [1]; the historical separated shortcut is not the proposed method in these comparisons.

1.1 Matched deletion and training protocol

Each condition uses the same 200 original train/calibration/test assignments. In the deletion condition, only the specified observation is removed from its assigned partition. All other memberships are preserved; the entire remaining dataset is not repartitioned. Training transformations and the predictor are refitted as required. Each pair of methods shares the same fitted predictor and residual arrays. Original rf2 partitions contain 5,759 training, 384 calibration, and 1,536 test observations; Crime partitions contain 1,426, 95, and 381, respectively. Deletion reduces exactly one partition by one row.

Both datasets use a jointly fitted multi-output random forest with 200 trees, maximum depth 16, minimum split size 5, minimum leaf size 2, square-root feature subsampling, squared-error criterion, bootstrap sampling, and model seed 42. For standardized training, each target is centered and divided by its population standard deviation computed on the training partition alone. Predictions are transformed back to the original target units before residual calibration. This changes target weighting during fitting; it is not a post-fit rescaling of prediction regions. The retained rf2 condition required 200 new standardized fits; the removed condition reuses 200 previously completed standardized fits. Crime uses original and standardized training in both deletion conditions: 600 new fits and 200 reused original fits.

1.2 Regions, coverage, and volume

Let $\hat{f}$ be the location predictor, d the number of outputs, and $e_{ij} = |y_{ij} - \hat{f}_j(x_i)|$ an absolute residual. Both methods return nonnegative coordinate thresholds u_{bj}^M for split b and method M . Their outcome-space regions and full volumes are

$$C_b^M(x) = \prod_{j=1}^d [\hat{f}_j(x) - u_{bj}^M, \hat{f}_j(x) + u_{bj}^M], \quad V_b^M = \prod_{j=1}^d 2u_{bj}^M. \quad (1)$$

All experiments target $1 - \alpha = 0.90$ joint coverage: an observation is covered only if all coordinates are covered. Reported coverage is the arithmetic mean of the 200 split-level coverage estimates. Regions have constant volume within a split, although their centers depend on the test input. Volumes are products of *full interval lengths*; they are not the historical half-width products. Absolute volumes must not be compared across datasets with different units or output dimensions.

The primary relative-volume statistic is

$$R = \frac{\bar{V}^{\text{Env}}}{\bar{V}^{\text{CHR}}}, \quad \bar{V}^M = \frac{1}{200} \sum_{b=1}^{200} V_b^M. \quad (2)$$

A value below one favors envelope in mean volume. We also report the median of the 200 paired ratios $V_b^{\text{Env}}/V_b^{\text{CHR}}$ and the fraction of splits with a smaller envelope. These are different estimands and can lead to different assessments when volumes are highly skewed. Coverage changes are additionally evaluated on the identical test observations common to each pair, excluding the removed observation from both denominators.

2 rf2: strong sensitivity without a ranking reversal

The retained complete-case rf2 cohort has 7,679 observations and eight outputs. The removed observation has NASI2 target 78.9, at retained position 3660 and original dataframe index 4782 (zero-based indexing). The next largest value is 5.62; the median is 3.30 and the median absolute deviation (MAD) is 0.07. Its modified z-score, $0.67449(y - \text{median}(y))/\text{MAD}(y)$, is approximately 728.45. Neighboring original rows have values 3.56, 3.53, 3.53, 78.9, 3.54, 3.51, and 3.51. The original ARFF contains the value, so it was not introduced by the experiment cache. These observations establish statistical extremeness, without proving a measurement or entry error.

Table 1: rf2 with training-target standardization in both conditions. Coverage is in percent; coverage SD is across splits, in percentage points. Each row summarizes 200 splits.

Observation	Method	Coverage	Cov. SD	Mean volume	Median volume
Retained	Envelope	90.052	1.715	101,671.0	24,189.5
Retained	Point CHR	90.091	2.366	16,019.1	8,790.5
Removed	Envelope	90.068	1.679	14,691.5	10,197.1
Removed	Point CHR	90.017	2.402	9,944.2	5,654.3

Table 1 shows an 85.55% reduction in envelope mean volume, compared with 37.92% for Point CHR. The median within-split envelope reduction is 46.44%, which is smaller because the mean is more affected by extreme-volume splits. The mean-volume ratio falls from 6.347 to 1.477. Envelope is smaller in 45/200 retained splits and 52/200 removed splits. Thus, the outlier explains substantial inefficiency but does

not fully explain the remaining disadvantage. On common test observations, envelope coverage changes by only +0.0059 percentage points, with a descriptive 95% Monte Carlo interval $[-0.1305, +0.1422]$ percentage points.

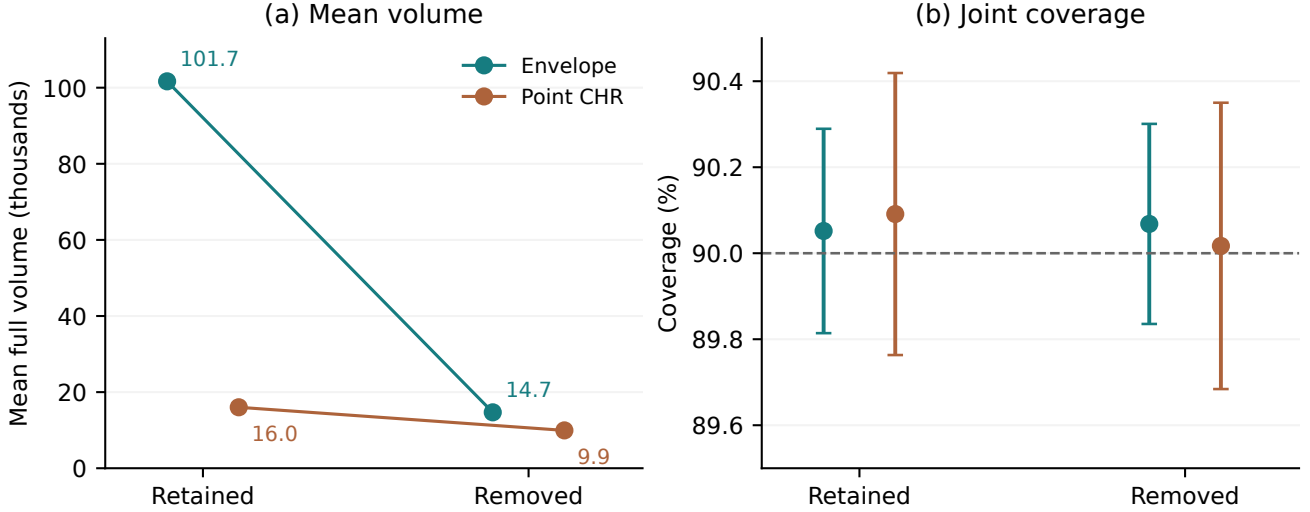


Figure 1: rf2 under standardized training. Panel (a) shows arithmetic mean full volume; panel (b) shows mean joint coverage with $1.96 \text{ SD} / \sqrt{200}$ Monte Carlo bars and the 90% target. The bars describe split variability conditional on the dataset, not uncertainty for an independently sampled population.

Training standardization changes the forest’s loss weighting but does not remove the effect of an extreme residual on calibration means and standard deviations. In the six splits placing the observation in calibration, envelope mean volume falls from 1,143,702 to 8,622 after deletion. In all 36 test-only deletions, the training and calibration data are unchanged and the bounds reproduce identically. Section 4 separates these roles.

rf2 consists of temporally dependent river-flow observations [2]. The controlled random-split comparison remains informative about sensitivity under that protocol. It does not establish coverage for a future forecasting task, where temporal separation and distribution shift require separate evaluation. A clear outlier-sensitivity effect and a failure to reverse the method ranking can therefore coexist.

3 Crime: a qualified reversal in mean volume

Crime contains community-level records rather than an hourly forecasting sequence [3]. The current cohort has 1,902 rows and 18 outcomes, combining eight crime counts and ten population-based rates. The removal rule was fixed before examining deletion outcomes: remove the community with the maximum input population. This is retained position 16, original dataframe index 21, with population 7,322,564 versus 3,485,398 for the next largest community. No iterative deletion or threshold search was used to obtain a win.

With standardized training, deleting this community reduces envelope’s mean volume from 7.0729×10^{65} to 5.8670×10^{62} , a 99.917% reduction. The corresponding Point CHR reduction is 68.34%. The ratio of means changes from 38.615 to 0.101, with envelope coverage remaining close to 90.65%. Original target training also exhibits a numerical reversal, from 2,081.97 to 0.0354, although CHR’s mean volume increases after deletion in that condition. There is no general monotonicity guarantee for volume after refitting and recalibrating on a reduced dataset.

Table 2: Crime under original and standardized target training. All entries use 200 matched splits and the corrected Point CHR recalibration rank. Coverage is in percent; volumes use all 18 full interval lengths.

Training targets	Community	Method	Coverage	Mean volume	Median volume
Original	Retained	Envelope	90.765	1.521×10^{66}	1.379×10^{57}
Original	Retained	Point CHR	91.706	7.304×10^{62}	1.247×10^{57}
Original	Removed	Envelope	90.803	5.406×10^{62}	8.106×10^{56}
Original	Removed	Point CHR	91.665	1.529×10^{64}	5.178×10^{56}
Standardized	Retained	Envelope	90.655	7.073×10^{65}	1.086×10^{57}
Standardized	Retained	Point CHR	91.470	1.832×10^{64}	8.265×10^{56}
Standardized	Removed	Envelope	90.647	5.867×10^{62}	4.795×10^{56}
Standardized	Removed	Point CHR	91.562	5.800×10^{63}	5.329×10^{56}

Table 3: Crime ranking diagnostics. R is the ratio of mean volumes from Equation (2); the median is over paired split-level ratios. The final column is a descriptive 95% paired split-bootstrap interval for R .

Training	Community	R	Paired median	Env. smaller	Interval for R
Original	Retained	2,082	1.104	100/200	$[14.3, 1.14 \times 10^4]$
Original	Removed	0.03535	0.906	102/200	$[0.00516, 4.29]$
Standardized	Retained	38.61	1.793	94/200	$[0.538, 2.85 \times 10^4]$
Standardized	Removed	0.1012	0.932	100/200	$[0.0113, 2.58]$

The mean reversal is less persuasive as evidence of a typical-split advantage. After standardized deletion, envelope is smaller in exactly 100/200 splits, its median paired ratio is 0.932, and its geometric mean paired ratio is 1.084. Furthermore, the 95% bootstrap interval for R is $[0.0113, 2.58]$, crossing one. Both the empirical mean and these uncertainty estimates are sensitive to the extreme observed splits. Consequently, Table 2 supports a numerical mean-volume reversal in this experiment, not a reliable general superiority claim.

4 Mechanism, influential tails, and coverage

For calibration residual coordinate j , envelope uses statistics of the form

$$m_j = \frac{1}{n} \sum_{i=1}^n e_{ij}, \quad s_j^2 = \frac{1}{n} \sum_{i=1}^n (e_{ij} - m_j)^2, \quad (3)$$

within its candidate-augmented surface construction [1]. An extreme residual can affect both statistics even when the forest’s targets were standardized during training. Because Equation (1) multiplies widths across coordinates, moderate coordinate inflation can become a large volume effect in 18 dimensions. This explains a plausible mechanism; it is not a claim that every extreme residual necessarily increases every width.

For Crime, only 11/200 splits place the largest community in calibration, but these account for 99.90% of the retained standardized envelope’s total volume. Their mean volume drops from 1.285×10^{67} to 1.092×10^{60} on deletion. The five largest retained-envelope trials contribute 99.28% of total volume. In contrast, the median paired envelope volume reduction from deletion is only 0.274%. Figure 3 makes this difference between average and typical behavior visible.

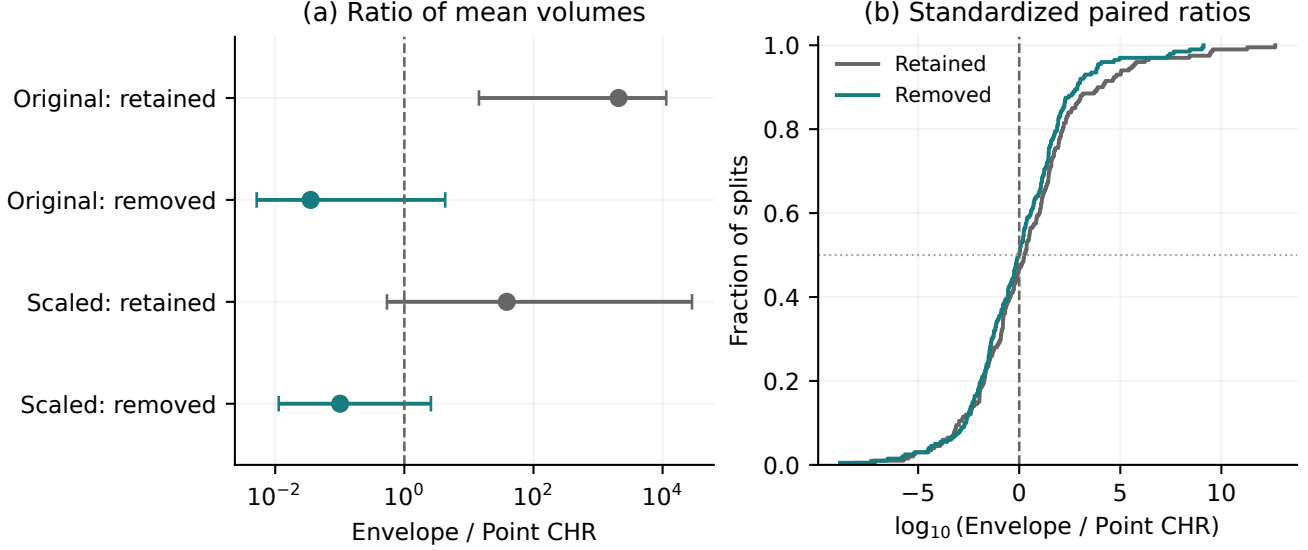


Figure 2: Crime mean and paired-ratio diagnostics. Panel (a) shows R with 20,000 paired split-bootstrap percentile intervals on a logarithmic axis. Panel (b) shows empirical cumulative distributions of standardized split-level log ratios. Ratios below one (log ratios below zero) favor envelope. The apparent mean advantage after deletion is much larger than the typical-split advantage.

Table 4: Envelope mean volume by the original partition of the removed row. Both datasets use standardized target training. Test-only deletion leaves training and calibration unchanged, providing an exact control.

Dataset	Original partition	Splits	Retained mean volume	Removed mean volume
rf2	Training	158	81,887.1	14,890.8
rf2	Calibration	6	1,143,702.3	8,621.9
rf2	Testing	36	14,828.6	14,828.6
Crime	Training	143	4.902×10^{62}	3.336×10^{62}
Crime	Calibration	11	1.285×10^{67}	1.092×10^{60}
Crime	Testing	46	1.513×10^{63}	1.513×10^{63}

Table 5: Paired deletion effects with standardized training. Reductions are percentages; coverage changes and their Monte Carlo intervals are percentage points on the identical remaining test observations. Positive reductions indicate smaller regions after deletion.

Dataset	Method	Volume reduction (%)		Common-test coverage	
		Mean	Paired median	Change	95% interval
rf2	Envelope	85.550	46.444	+0.0059	[-0.130, +0.142]
rf2	Point CHR	37.923	17.165	-0.0846	[-0.286, +0.116]
Crime	Envelope	99.917	0.274	-0.0630	[-0.252, +0.126]
Crime	Point CHR	68.337	0.000	+0.0367	[-0.216, +0.290]

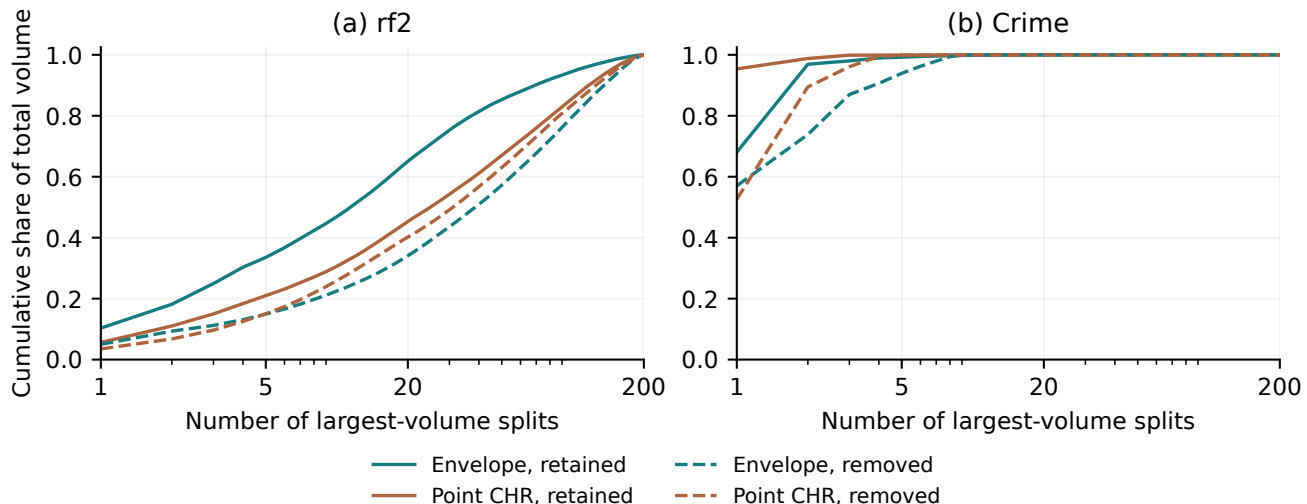


Figure 3: Concentration of volume across the 200 splits, with standardized training. Each curve sorts its own split volumes from largest to smallest and plots their cumulative share of the total. Rapid saturation means that few splits dominate the arithmetic mean. The horizontal axis is logarithmic; the split ordering can differ between methods and conditions.

5 Outlier status and relevance of other cohorts

The rf2 observation is exceptionally isolated relative to the rest of its target distribution. Crime’s largest community is better described as an influential upper-tail observation. Its maximum robust z-score across $\log(1 + y)$ -transformed targets is 6.696, followed by 6.405 and 6.120 for the next two observations under the same statistic. These transformed scores are descriptive and should not be compared numerically with rf2’s raw-target z-score as a universal severity scale.

In Crime, the community has the largest value for seven of the eight count outcomes, but its population-based rates are less exceptional. For example, its larceny count is 235,132, the cohort maximum, whereas its larceny rate is at approximately the 54th empirical percentile. Its nonviolent-crime rate is at approximately the 74th percentile. This is consistent with a city-size effect rather than an isolated spike across all outcomes. It does not verify the record’s accuracy. Removing a legitimate large community changes the benchmark population, so the analysis is labeled a largest-community sensitivity experiment rather than a corrected-data experiment.

Only rf2 and Crime start with larger envelope mean volume in this screen. The remaining finite-baseline cohorts already favor envelope on that metric. Energy is generated from building simulations; Student contains student-level records; Air is an hourly time series. Stock has an unbounded corrected CHR region in all 200 splits because its calibration halves are too small for the required finite conformal quantile. These facts limit their suitability for the requested finite-baseline loss-to-win example. The screen is descriptive: deletion fits were run for rf2 and Crime, not for all eight cohorts or every possible removal rule.

5.1 Interpretation for reporting the experiments

The strongest supported rf2 statement is that a single statistically extreme observation substantially inflates envelope’s mean volume even with training-target standardization, while its removal leaves coverage almost unchanged and does not reverse the CHR comparison. Crime provides the requested reversal in empirical mean volume, but the gain is concentrated in rare calibration-tail cases and is not established as a consistent advantage across splits. These outcomes support a focused claim about

Table 6: Screen of the eight existing cohorts under the complete-data, original-training benchmark at nominal 90% coverage. N is the retained cohort size and d the number of outputs. These ratios precede the new standardized controls; they must not be mixed with the standardized result tables.

Dataset	N	d	Env./CHR mean volume	Assessment
stock	315	6	Unbounded CHR	CHR is unbounded with the small calibration sample.
rf2	7,679	8	6.954	Isolated extreme value; no standardized reversal.
scm1d	9,803	16	0.535	Envelope already has smaller mean volume.
scm20d	8,966	16	0.508	Envelope already has smaller mean volume.
energy	768	2	0.776	Envelope already smaller; building simulations [4].
student	649	3	0.282	Envelope already smaller; student records [5].
air	6,941	4	0.983	Envelope slightly smaller; hourly series [6].
crime	1,902	18	2,081.972	Community records; fixed deletion tested [3].

sensitivity to influential observations. They do not support a general rule that outlier removal makes envelope outperform Point CHR, or that deletion is warranted in the primary benchmark.

A Calibration details, uncertainty, and reproducibility

A.1 Point CHR and the corrected order statistic

For a calibration multiset $v = (v_1, \dots, v_m)$, define

$$Q_{1-\alpha}(v) = \begin{cases} v_{(r)}, & r = \lceil (m+1)(1-\alpha) \rceil \leq m, \\ +\infty, & r > m. \end{cases} \quad (4)$$

Point CHR randomly divides the calibration residuals into two halves, A and B , using seed 42. With $q_j = Q_{1-\alpha}(\{e_{ij} : i \in A\})$, its thresholds are

$$u_j^{\text{CHR}} = q_j Q_{1-\alpha} \left(\left\{ \max_k \frac{e_{ik}}{q_k} : i \in B \right\} \right). \quad (5)$$

Each quantile uses its own sample size. The original 95-row Crime calibration set gives halves of 47 and 48 observations; deleting a calibration row gives 47 and 47. The original rf2 calibration halves are 192 and 192; calibration-row deletion gives 191 and 192. The same seeded splitting procedure is rerun on the reduced calibration array, so internal CHR half membership need not remain identical after deletion. This is part of applying the fixed procedure to the reduced data; outer train/calibration/test memberships remain matched.

The current analysis consistently uses the corrected second-half recalibration rank. Historical raw Crime archives preserve pre-correction CHR bounds; their residual arrays, rather than their obsolete CHR bounds, are reused for this comparison. No alternative rank was selected to improve envelope's ranking.

A.2 Monte Carlo summaries and their limits

Coverage bars and paired coverage-change intervals use split-level standard deviations and the normal approximation $1.96 \text{SD} / \sqrt{200}$. Crime mean-volume intervals use 20,000 bootstrap samples of the 200 split indices with replacement, random seed 20260909, retaining method and treatment pairing. The 2.5th and 97.5th percentiles of the bootstrap ratios form the reported intervals. All such uncertainty summaries are conditional on the fixed dataset and random-split protocol. They do not represent uncertainty from drawing a new population sample, account for cohort selection, or resolve the effects of rare extreme splits not observed among the 200 trials.

rf2's row was identified after examining its outcomes, and Crime was investigated after screening existing method results. These are exploratory sensitivity analyses, not preregistered confirmatory tests. The Crime deletion rule was fixed before its deletion results were computed. The nominal 90% level is a target; near-nominal empirical coverage under these splits does not by itself establish exchangeability or validate forecasting claims.

A.3 Completed work and audit trail

The rf2 audit verifies original source hashes, exact deletion alignment, training-only centers and scales, retained method metrics, and independently recomputed removed-condition metrics. All 36 test-only deletion pairs reproduce their bounds. Crime contains 800 fitted conditions (600 new and 200 reused). Its fit audit checks deletion membership and training transformations; its reporting audit recomputes calibration bounds and all 1,600 method coverage and volume records. Across 46 test-only deletion splits, two training choices, and two methods, all 184 bound comparisons reproduce exactly. Source and archive checksums are retained. Source observations and primary benchmark results were not overwritten. No additional model fitting was performed to prepare this report.

The portable report folder contains the complete `.tex` source, vector PDF figures, the compiled report, selected CSV/JSON evidence under `data/`, and a manifest of source checksums. Full fitted residual archives remain in the project. The source locations relative to the project root are:

- `envelope_method/results/rf2_standardized_outlier_control/`
- `envelope_method/results/rf2_remaining/model_time/`
- `envelope_method/results/real_outlier_screen/crime_control/`
- `envelope_method/results/real_comparison_audited.csv`

The report builder is `envelope_method/build_outlier_sensitivity_report.py`. It verifies summary statistics against trial data and regenerates tables and figures. The report source can be compiled from its folder with `pdflatex` twice, or with `tectonic`.

References

- [1] Project method note. *Signed Surface-Envelope Transductively Standardized Conformal Prediction: Construction, finite-sample proofs, and comparison with the separated shortcut*. September 8, 2026. Source: `envelope_method/signed_envelope.tex`.
- [2] Mulan project. *Multi-target regression datasets*, including River Flow 2.
<https://mulan.sourceforge.net/datasets-mtr.html>.
- [3] UCI Machine Learning Repository. *Communities and Crime Unnormalized*, dataset 211.
<https://archive.ics.uci.edu/dataset/211/communities+and+crime+unnormalized>.
- [4] UCI Machine Learning Repository. *Energy Efficiency*, dataset 242.
<https://archive.ics.uci.edu/dataset/242/energy+efficiency>.
- [5] UCI Machine Learning Repository. *Student Performance*, dataset 320.
<https://archive.ics.uci.edu/dataset/320/student+performance>.
- [6] UCI Machine Learning Repository. *Air Quality*, dataset 360.
<https://archive.ics.uci.edu/dataset/360/air+quality>.